

AURUS V2 - Intelligent Home Companion Rework Specification

Project Vision

AURUS V2 shall evolve from a voice-controlled robotic pet into an emotionally intelligent autonomous home companion capable of:

- Recognizing and interacting with users
- Maintaining long-term memory
- Tracking user presence
- Following users visually
- Demonstrating contextual awareness
- Exhibiting adaptive personality traits
- Providing meaningful daily interactions

The project must continue using the existing hardware platform:

Hardware Constraints

- Raspberry Pi 4B (8GB)
- Raspberry Pi Camera V2
- 5x HC-SR04 Ultrasonic Sensors
- 4x Mecanum Wheels
- 2x L298N Drivers
- USB Microphone
- Speakers

No additional hardware shall be added.

SYSTEM ARCHITECTURE

Companion Brain

The system shall be divided into separate modules:

Voice Module

Responsibilities:

- Wake word detection
- Speech transcription

- Intent extraction

Vision Module

Responsibilities:

- Face detection
- User tracking
- Presence monitoring
- Object recognition

Memory Module

Responsibilities:

- Persistent user memories
- Event logging
- Interaction history
- Preference storage

Emotion Module

Responsibilities:

- Emotional state calculations
- Relationship tracking
- Social engagement analysis

Companion Core

Responsibilities:

- Context awareness
- Decision making
- Behavior scheduling
- Task execution

Mobility Module

Responsibilities:

- Following
- Navigation
- Obstacle avoidance
- Expressive movement

TECH STACK REWORK

Remove

SpeechRecognition

Reason:

- Slow
 - Unreliable
 - High latency
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Add

Faster Whisper

Purpose:

Local speech-to-text processing.

Advantages:

- Better accuracy
 - Faster processing
 - Offline operation
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Porcupine

Purpose:

Wake word detection.

Wake Phrase:

"AURUS"

Advantages:

- Low CPU usage
 - Offline operation
 - Raspberry Pi optimized
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OpenCV

Purpose:

Computer vision foundation.

Functions:

- Camera processing
 - Face tracking
 - Motion detection
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MediaPipe

Purpose:

Human tracking.

Functions:

- Face detection
 - Landmark tracking
 - User positioning
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SQLite

Purpose:

Persistent memory storage.

Tables:

Users Memories Events Interactions Preferences

Gemini API

Purpose:

Conversation Reasoning Personality responses

Gemini shall NOT control:

- Motor functions

- Navigation
- Basic commands

Those actions shall execute locally.

UI REDESIGN

Current dashboard shall be redesigned into five sections.

SECTION 1 - Companion Status

Display:

- Current mood
- Current activity
- Battery status
- Uptime
- Active mode

Examples:

Idle Following User Listening Thinking Exploring

SECTION 2 - Vision Feed

Live camera feed.

Display overlays:

- Face bounding boxes
 - User labels
 - Tracking indicators
 - Detected objects
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SECTION 3 - Emotional Analytics

Charts displaying:

- Happiness
- Curiosity
- Trust

- Social Energy

Historical graph:

Last 24 hours

SECTION 4 - Memory Center

Display:

Recent memories

Examples:

"User discussed robotics project."

"User returned home."

"User requested follow mode."

SECTION 5 - Explainable AI Panel

Every autonomous action must display:

Decision: Follow User

Reason: User detected and follow mode enabled.

Confidence: 94%

This demonstrates explainable AI concepts.

PHASE 1 - CORE PERFORMANCE REWORK

Priority: CRITICAL

Goal:

Reduce latency and improve responsiveness.

Tasks:

1. Replace SpeechRecognition with Faster Whisper.
2. Replace wake word logic with Porcupine.
3. Build Intent Router.

Intent Router must classify:

Movement Commands Follow Commands Mode Changes Conversation Requests

Example:

"Move forward"

Should execute locally.

Gemini must not be called.

Acceptance Criteria:

Voice command response under 1 second.

PHASE 2 - COMPANION MEMORY SYSTEM

Priority: CRITICAL

Goal:

Give AURUS continuity.

Tasks:

Create SQLite database.

Implement:

Users Table

Memories Table

Events Table

Preferences Table

Interactions Table

Memory Types:

Conversation Memory

Preference Memory

Event Memory

Behavior Memory

Examples:

User likes science fiction.

User works on robotics.

User returned at 6PM.

Acceptance Criteria:

AURUS recalls previous conversations.

PHASE 3 - PRESENCE AWARENESS

Priority: HIGH

Goal:

Allow AURUS to understand human presence.

Tasks:

Implement:

Face Detection

Motion Detection

Presence Detection

User State Tracking

Presence States:

Present

Nearby

Away

Absent

Example:

Human enters room.

AURUS greets user.

Acceptance Criteria:

AURUS can determine whether a user is present.

PHASE 4 - FOLLOW MODE

Priority: HIGH

Goal:

Create purposeful mobility.

Tasks:

Implement visual following.

Pipeline:

Camera → Face Detection → Position Estimation → Mecanum Controller

Behavior:

Maintain approximately 1 meter distance.

Stop when user stops.

Avoid obstacles using ultrasonic sensors.

Acceptance Criteria:

AURUS follows user safely.

PHASE 5 - ADAPTIVE EMOTIONAL ENGINE

Priority: HIGH

Goal:

Transform mood system into relationship system.

Replace:

Fear Curiosity Happiness

With:

Happiness

Trust

Social Energy

Curiosity

Relationship Strength

Triggers:

Positive Interaction

Ignored Interaction

User Presence

Conversation Frequency

Acceptance Criteria:

Emotional state changes based on interaction history.

PHASE 6 - COMPANION INITIATIVE SYSTEM

Priority: MEDIUM

Goal:

Allow AURUS to initiate interactions.

Examples:

"Good morning."

"I haven't seen you today."

"You've been studying for a while."

Behavior Types:

Greeting

Check-In

Encouragement

Idle Conversation

Acceptance Criteria:

AURUS starts conversations naturally.

PHASE 7 - OBJECT RECOGNITION

Priority: MEDIUM

Goal:

Improve environmental awareness.

Objects:

Laptop

Phone

Bottle

Chair

Person

Book

Examples:

"I can see a laptop."

"You appear to be working."

Acceptance Criteria:

Object detection visible on dashboard.

PHASE 8 - DAILY COMPANION REPORT

Priority: MEDIUM

Goal:

Demonstrate memory and analytics.

Command:

"AURUS summarize today."

Output:

Interactions count

Presence duration

Major activities

Notable events

Generated using Gemini.

Acceptance Criteria:

Meaningful daily summary generated.

PHASE 9 - EXPLAINABLE AI SYSTEM

Priority: LOW

Goal:

Provide transparency.

Every autonomous action must include:

Decision

Reason

Confidence

Source Data

Displayed on dashboard.

Acceptance Criteria:

All actions can be explained.

PHASE 10 - DEMONSTRATION MODE

Priority: FINAL

Goal:

Create evaluation-ready workflow.

Demo Sequence:

1. User enters room.
2. AURUS recognizes user.

3. AURUS greets user.
4. User asks question.
5. AURUS remembers previous interaction.
6. User activates follow mode.
7. AURUS follows user.
8. User leaves room.
9. AURUS detects absence.
10. User requests daily summary.

This demonstration shall showcase:

AI

Computer Vision

Memory

Sensor Fusion

Autonomous Navigation

Human-Robot Interaction

Explainable AI

Database Integration

Embedded Systems